



Material Declaration Sheet

Draloric/Beyschlag Resistor

Date 12/2/2013

Part / Product Family Details

Vishay Part Number	RoHS Compliance Status	RoHS Compliance Date Code dd-mm-yyyy	Total product Weight (mg)	Value Range	3rd Party Lab ICP Test Report Available	Manufacturing Location	Number of Exemptions Used
RCA120600002SEA	YES WITH EXEMPTION	Since April 2005	8.82484	OR	Yes	Germany	Three

Technical Information

Terminal Plating / Grid Array Material	Terminal Base Alloy	JESD-97 Pb-Free Material Code Marking	J-STD-20D MSL Rating	Reflow Peak Process Body Temperature	Reflow Maximum number of cycles	Reflow Max.Time at Peak Temperature (sec)	Soldering Compatibility (SnPb/Pb-Free)
Tin (Sn)	Other	e3	1	260 ° C	3	40	Backward & Forward

Material Composition

Homogenous Material Name	Material Classification	Substance Name	CAS number	Weight of Substance (mg)	With respect to Homogenous Material		% with respect to Total Product Weight	RoHS Exemptions Used
					%	ppm		
Ceramic Substrate				7.95703				
	Magnesium Oxide	Magnesium Oxide	1309-48-4	0.07878	0.99	9901	0.89	0 - None
	Aluminium Oxide	Aluminium Oxide	1344-28-1	7.61010	95.64	956400	86.23	0 - None
	Silicon Dioxide	Silicon Dioxide	7631-86-9	0.23155	2.91	29100	2.62	0 - None
	Miscellaneous	Miscellaneous	System	0.03660	0.46	4600	0.41	0 - None
Passivation				0.04545				
	Lead(2+) Silicate	Lead(2+) Silicate	10099-76-0	0.04450	97.91	979098	0.50	Exemption No: -7(c)-I
	Chromium (III) Oxide	Chromium (III) Oxide	1308-38-9	0.00095	2.09	20902	0.01	0 - None
Resistive layer				0.03084				
	Lead(2+) Silicate	Lead(2+) Silicate	10099-76-0	0.01322	42.87	428664	0.15	Exemption No: -7(c)-I
	Silver	Silver	7440-22-4	0.01762	57.13	571336	0.20	0 - None
Termination				0.65207				
	Lead(2+) Silicate	Lead(2+) Silicate	10099-76-0	0.00460	0.71	7054	0.05	Exemption No: -7(c)-I
	Copper Oxide (Derivatives-1)	Copper Oxide (Derivatives-1)	1317-38-0	0.00576	0.88	8833	0.07	0 - None
	Nickel	Nickel	7440-02-0	0.14766	22.64	226450	1.67	0 - None
	Silver	Silver	7440-22-4	0.21981	33.71	337096	2.49	0 - None
	Tin	Tin	7440-31-5	0.27423	42.06	420553	3.11	0 - None
	Chromium	Chromium	7440-47-3	0.00000	0.00	6	0.00	0 - None
	Copper	Copper	7440-50-8	0.00001	0.00	8	0.00	0 - None
Top Coat, Marking				0.13945				
	Epoxy Resine	Epoxy Resine	25068-38-6	0.01725	12.37	123667	0.20	0 - None
	Silicon Dioxide	Silicon Dioxide	7631-86-9	0.12220	87.63	876333	1.38	0 - None

EU-RoHS Directive-2011/65/EU MCV of 0.1% by mass (1000 PPM) in homogeneous material for: Lead (Pb), Mercury, Hexavalent Chromium, Polybrominated Biphenyls (PBB), Polybrominated Diphenyl Ethers (PBDE) and MCV of 0.01% by mass cadmium

Exemption Used 7(c)-I Electrical and electronic components containing lead in a glass or ceramic other than dielectric ceramic in capacitors, e.g. piezoelectronic devices, or in a glass or ceramic matrix compound

2nd Exemption Used 7(c)-I Electrical and electronic components containing lead in a glass or ceramic other than dielectric ceramic in capacitors, e.g. piezoelectronic devices, or in a glass or ceramic matrix compound

3rd Exemption Used 7(c)-I Electrical and electronic components containing lead in a glass or ceramic other than dielectric ceramic in capacitors, e.g. piezoelectronic devices, or in a glass or ceramic matrix compound

Note :- (i) All information is based on data received from our vendors & subjected to change without prior notice.
(ii) Substance weight are derived from MSDS.

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